SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

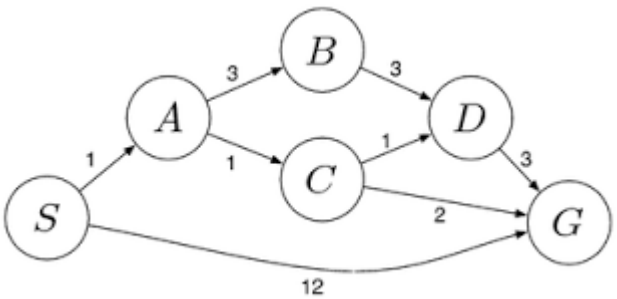
1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.
   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANALYTICAL PROBLEM SOLVING – ASSESSMENT

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Course / Subject: Artificial Intelligence

1. Water Jug Problem

Given: A 4-gallon jug and a 3-gallon jug are available. Both jugs have no measurement markings.

Goal: Measure exactly 2 gallons of water in the 4-gallon jug.

State Representation: A state is represented as (amount of water in 4-gallon jug, amount of water in 3-gallon jug).

Solution:

1. Fill the 3-gallon jug.
2. Pour it into the 4-gallon jug.
3. Fill the 3-gallon jug again.
4. Pour it into the 4-gallon jug until the 4-gallon jug becomes full.
5. The 3-gallon jug will have 2 gallons remaining.
6. Empty the 4-gallon jug.
7. Transfer the remaining 2 gallons into the 4-gallon jug.

Result: Exactly 2 gallons of water are obtained in the 4-gallon jug.

2. Mars Rover – Intelligent Agent

i. **Percepts:** The rover observes rocks, soil, obstacles, temperature, atmosphere, distance, terrain, sample sites, and possible signs of water or minerals.

ii. **Operating Environment:** The Mars rover works in an environment that is partially observable, dynamic, sequential, uncertain, continuous, and mostly single-agent.

iii. **Actions:** The rover can move, rotate, collect and examine samples, capture images, avoid obstacles, navigate different areas, and send information to Earth.

iv. **Performance Measures:** Its performance can be evaluated based on the number of useful samples collected, accuracy of analysis, distance covered, time used, energy consumption, safe obstacle avoidance, scientific discoveries, and successful completion of the mission.

v. Suitable Architecture:

A Model-Based Utility-Based Agent is suitable. It keeps track of information about its surroundings and selects actions by considering useful goals such as collecting samples, maintaining safety, and saving energy.

3. 8-Queens Problem

Problem: The objective is to arrange 8 queens on an 8 × 8 chessboard so that no queen can attack another queen.

Initial State: The chessboard is completely empty.

State: A state represents a partial placement of queens on the board.

Actions: Place the next queen in a position where it cannot be attacked by any previously placed queen.

Transition Model: Placing a queen in a safe position creates a new valid state.

Goal Test: The solution is reached when all 8 queens are placed without any two queens sharing the same row, column, or diagonal.

Path Cost: Each queen placement is assigned a cost of 1. Therefore, placing all 8 queens gives a total cost of 8.

Suitable Search: Backtracking / Depth First Search (DFS) is suitable because an incorrect arrangement can be rejected immediately.

One Valid Solution: [1, 5, 8, 6, 3, 7, 2, 4]

Here, each number indicates the row position of the queen in each column from left to right.

4. OLA Cab Booking – Problem-Solving Agent

Problem: A passenger needs to travel from a starting location to a destination using an OLA cab service. The passenger can choose options such as Mini, Micro, Sedan, Prime, or Shared, depending on price, comfort, and travel needs.

Agent Type: Utility-Based Agent

Reason: The agent evaluates different cab options by considering parameters such as fare, journey time, comfort, and customer preferences. It then chooses the option that provides the greatest overall utility.

Pseudocode

START

1. Enter the source location.

2. Enter the destination location.

3. Search for available cabs.

4. For every available cab:

1. Find the estimated fare.

2. Calculate the expected travel time.

3. Check the comfort level.

4. Determine the utility value.

5. Choose the cab with the highest utility.

6. Show the selected cab, fare, and estimated travel time.

7. Ask the customer for confirmation.

8. If the customer accepts:

1. Confirm the cab booking.

2. Display the booking information.

9. Otherwise:

1. Cancel the booking.

STOP

5. Logistics Delivery Network – Uniform Cost Search

Given Edge Costs:

• S → A = 1

• S → G = 12

• A → B = 3

• A → C = 1

• B → D = 3

• C → D = 1

• C → G = 2

• D → G = 3

Uniform Cost Search (UCS): UCS always chooses and expands the path with the smallest accumulated cost.

Stage	Path / Node	Total Cost
Start	S	0
1	S → A	1
2	S → A → C	2
3	S → A → C → D	3
4	S → A → C → G	4

Comparison:

The route S → A → C → G has a total cost of:

1 + 1 + 2 = 4

The direct route S → G has a cost of 12. Therefore, UCS chooses the route with the lower total cost.

Final Answer:

Least-cost path = S → A → C → G

Minimum cost = 4